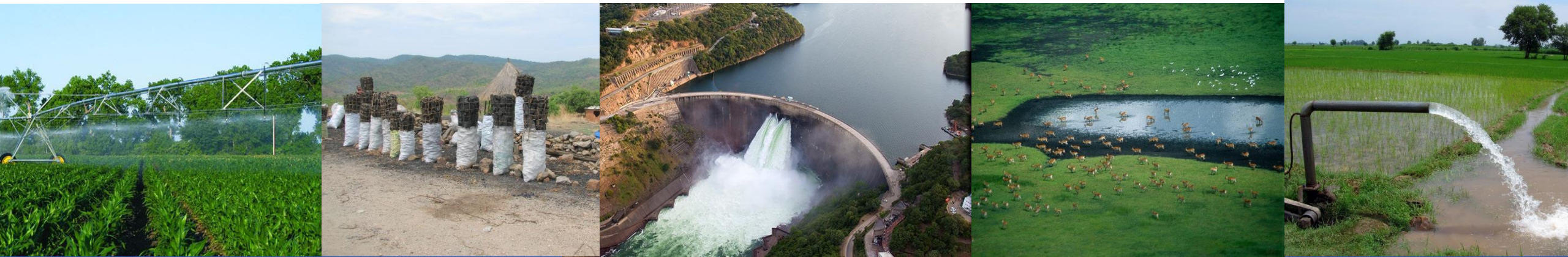


Exercise 8: Changing input data

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Changing input data

0. What you need
1. What is netCDF
2. Display netCDF files
3. Change ksat2 (saturated soil conductivity layer 2)
in ArcGIS



Changing input data

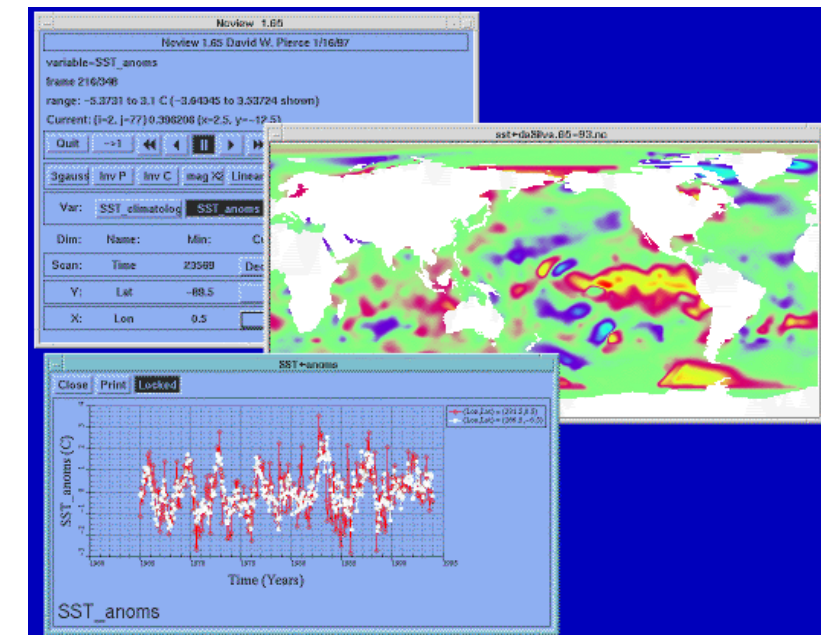
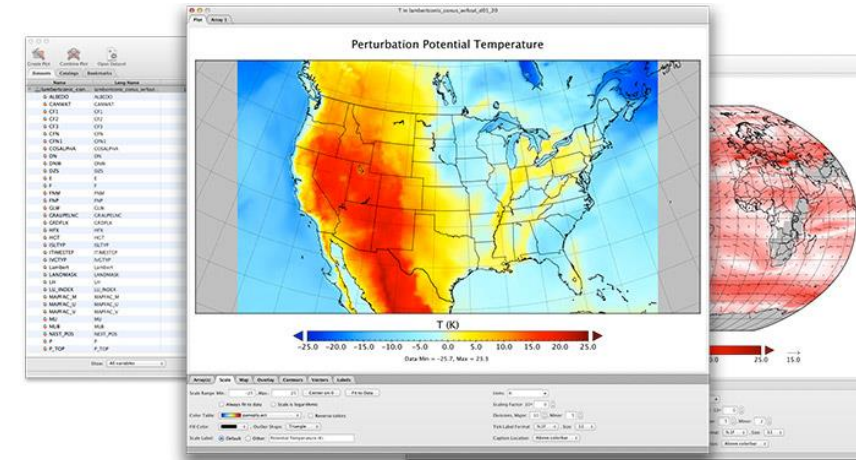
0. What you need

- A Geoinformation System e.g. QGIS, ArcGIS
- A way to display netCDF files easily e.g.:

PANOPLY on Windows

<https://www.giss.nasa.gov/tools/panoply/>

ncview on Linux:



Changing input data

What is netCDF

- **NetCDF (Network Common Data Form)** is a data format like .tif, .jpg
- **NetCDF** is a set of libraries and self-describing, machine-independent data formats that support the creation, access, and sharing of array-oriented scientific data
- Mainly used by the meteorological community

Advantage:

- netCDF is self-describing, portable, flexible
- netCDF is public domain, well documented and used by a growing number of organizations
- High efficiency in reading and processing netCDF files (it is fast)
- A netCDF file can contain timeseries of spatial data in a compressed way

Disadvantage:

- Not so easy to handle

Changing input data

Examples of netCDF

Digital elevation (source Danube: Merit DEM)

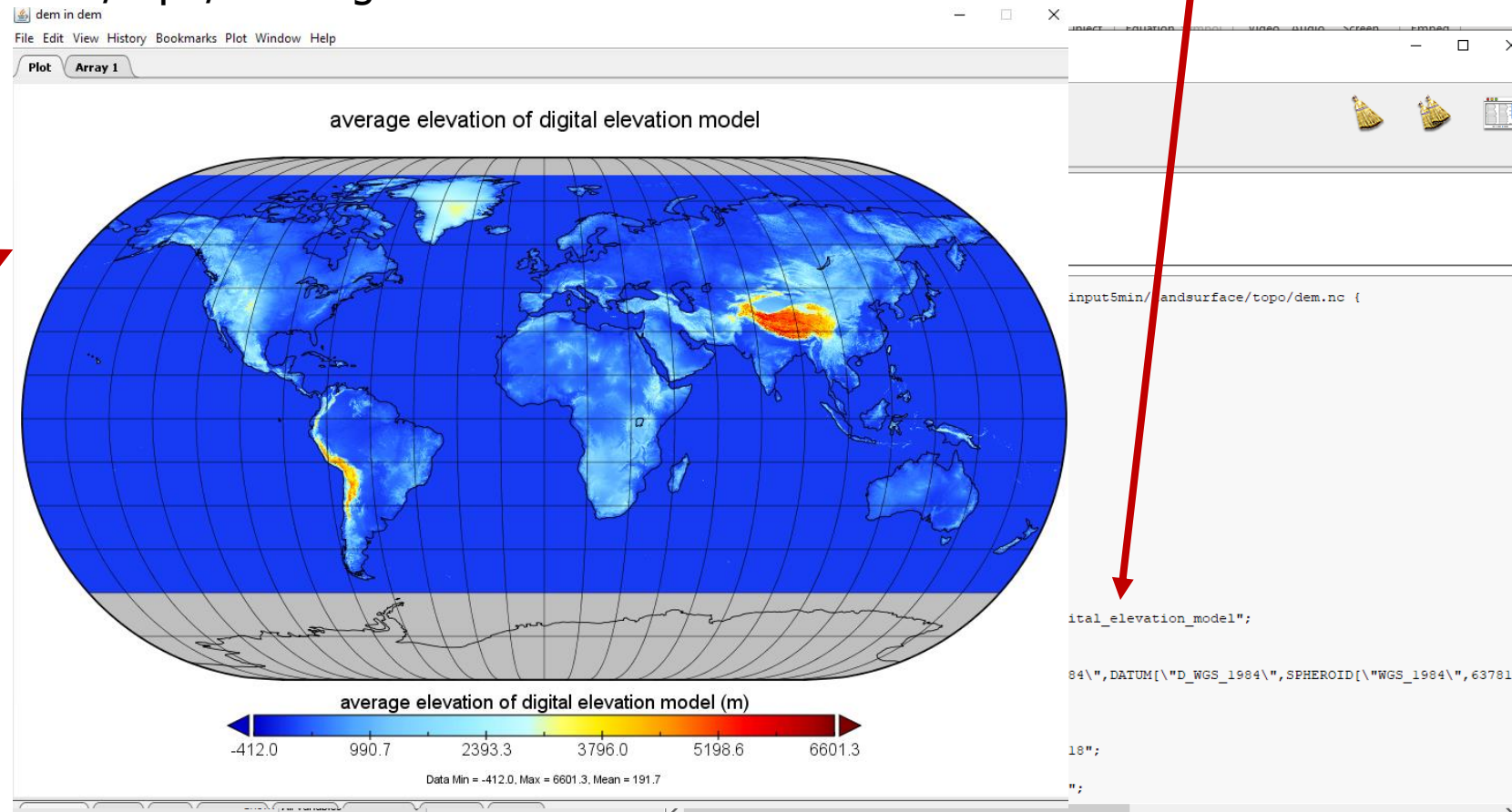
input/landsurface/topo/demavg.nc

Metainformation e.g.

- dimension (here without time)
- unit
- description

Variables

- latitude
- longitude
- elevation



Changing input data

Examples of netCDF

Monthly industry water demand (calculation Wada et al. 2014)

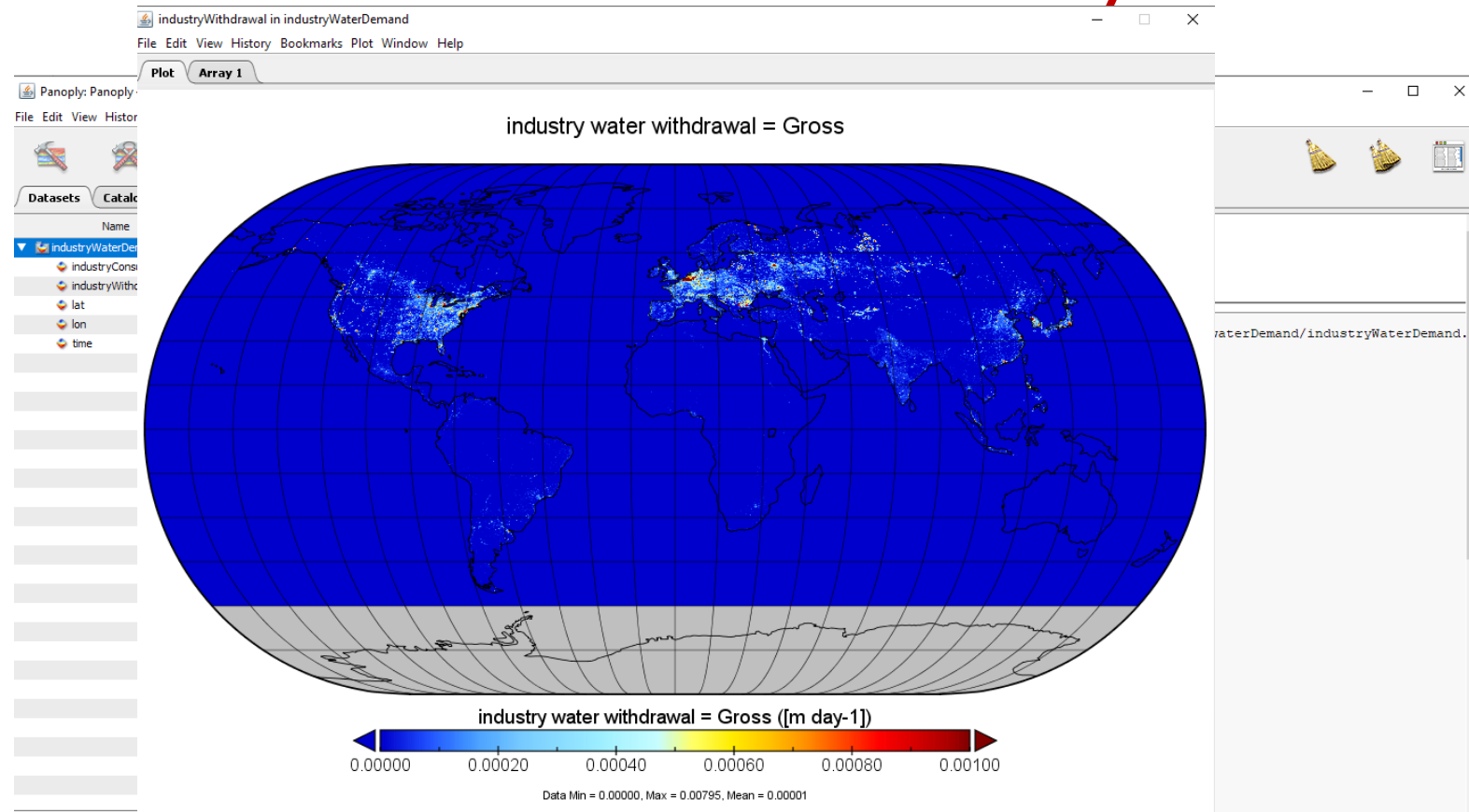
../CWATM_data/cwatm_input5min/landsurface/waterDemand/industryWaterDemand.nc

Metainformation e.g.

- dimension (here **with** time)
- unit
- description

Variables

- latitude
- longitude
- **time**
- **2 variables**

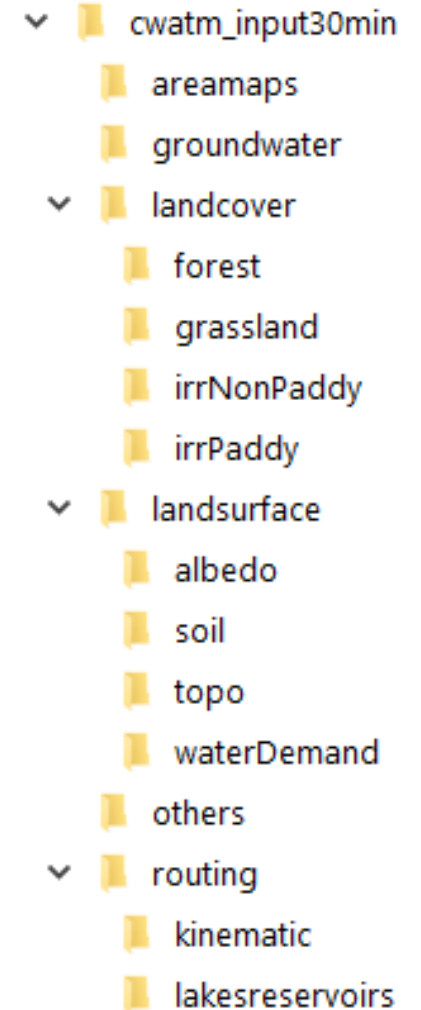


Changing input data

Take a look at dataset

- Take a look at:
<https://cwatm.iiasa.ac.at/data.html>
- Take a look at
./input

Data storage structure



Changing input data

Changing values with CDO (Linux)

CDO (Climate Data Operators)

<https://code.mpimet.mpg.de/projects/cdo/>

CDO is a collection of command line Operators to manipulate and analyse netCDF data.

There are more than 600 operators available

Metainformation e.g.

- dimension (here **with** time)
- unit
- description

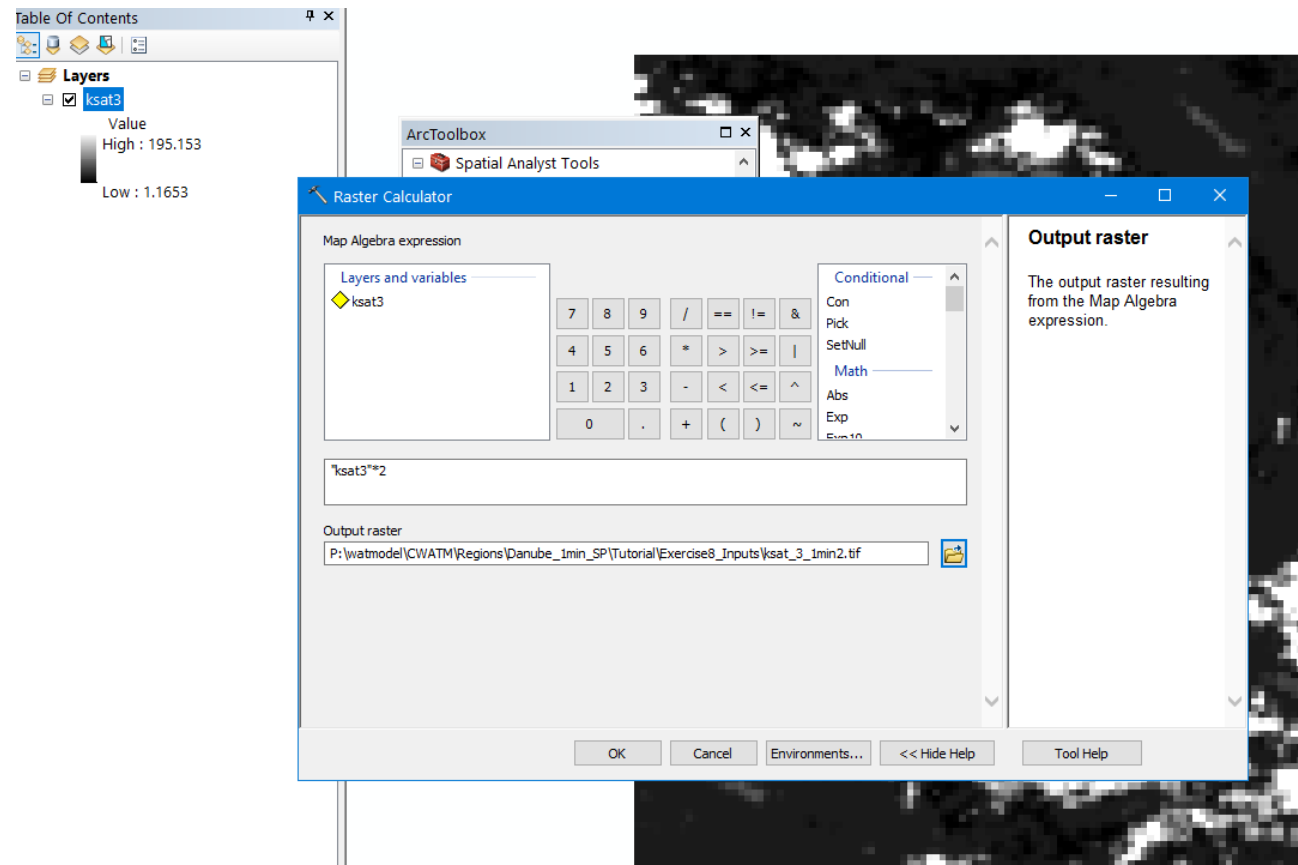
```
cdo expr, 'ksat2 = ksat2_o * 2' ksat2.nc ksat22.nc
```

Multiplying values by two

Changing input data

Changing values with ArcGIS

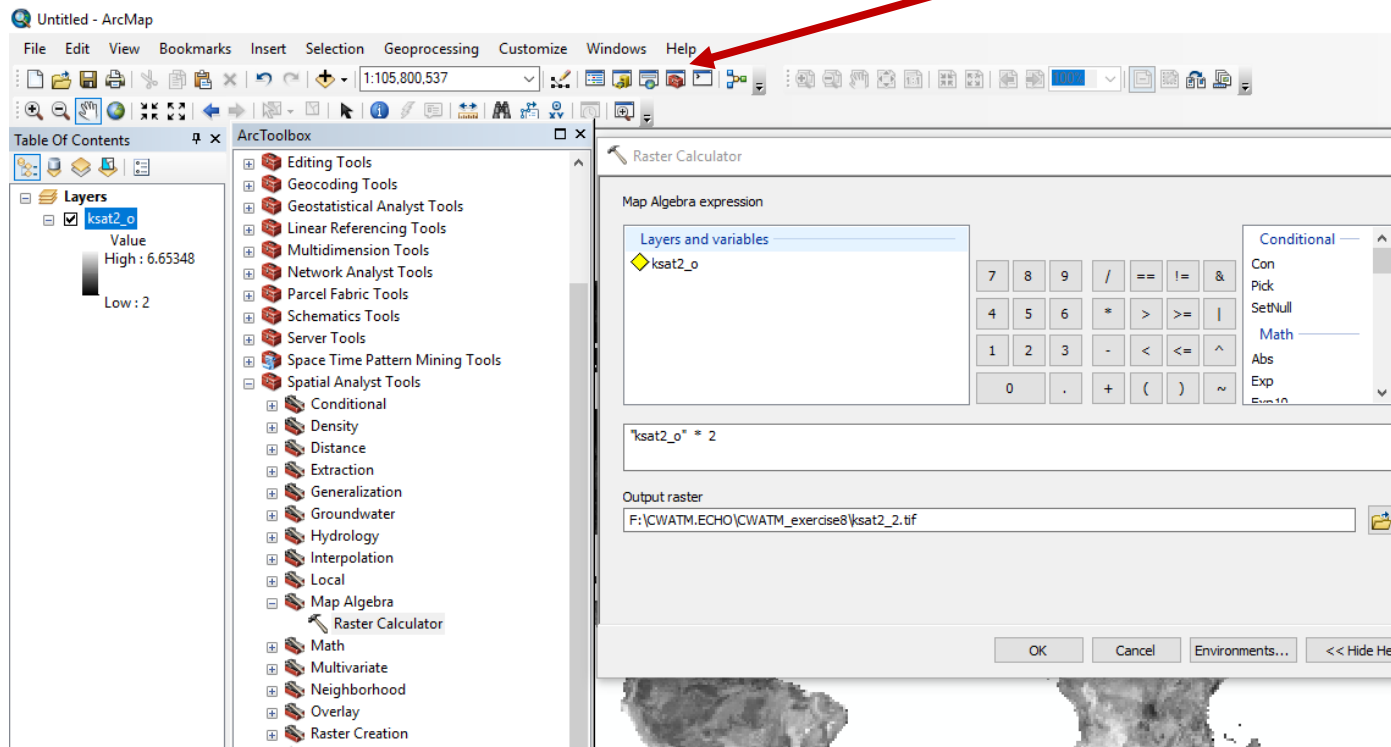
- Open ArcGIS
- Drop ksat2.nc into “Table of content”



Changing input data

Changing values with ArcGis

- Open toolbox – Spatial Analyst Tool – Map Algebra – Raster calc.
- Put in: "ksat2_o" *2
- Put in output raster – into the folder of CWATM_exercise8



You have to choose a different name than ksat2_2.tif if there is already a ksat2_2.tif in this folder

Changing input data

Convert file from .tif to netcdf

- Run tif2nc.bat
- This will run python tif2netcdf.py
- Output is ksats2_2.nc

```
115
116     tif2netcdf.py
117     # -----
118     # -----
119     # -----
120
121     inDir = "/"
122     outDir = "/"
123     netcdfOut = outDir + "ksats2_2.nc"
124     inName1 = inDir+"ksats2.tif"
125     timeattr = False
126
127     varShortNames = ['ksats2']
128     varLongNames = ['saturated soil conductivity layer2']
129     # We have to standardize units based on the CF convention.
130     varUnits = '[cm]'
131
```

Changing input data

Run CWatM twice to see the difference

- Run 81_exercise2.bat

Or `python ../../CWatM_model_developversion/run_cwatm.py settings_morava_1min_exercise81.ini -l-`

- Output in ./Exercise8_Inputs/out

- Copy results into another Excel

```
411  
412 #KSat2 = $(PathSoil)/ksat_3_1min.nc  
413 KSat2 = ./ksat_3_1min.nc  
414 #KSat2 = ./ksat_3_1min2.nc  
415
```

Difference in settings files

Changing input data

Put the results in Excel to see the difference

